

## PRACTICAL COURSE WORKBOOK

# Cloud Security - AWS & GCP

Apply practical protection to accounts and systems

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| LEVEL            | Intermediate                                                  |
| GUIDED TIME      | 5 hours across 12 lessons                                     |
| PROJECT          | a prioritised security review and action plan                 |
| SKILLS           | AWS, GCP, IAM, Zero Trust, Critical evaluation, Documentation |
| EDITORIAL STATUS | Structured draft v0.9                                         |

## WHAT YOU WILL BE ABLE TO DO

- Use AWS appropriately in a realistic, bounded task.
- Use GCP appropriately in a realistic, bounded task.
- Use IAM appropriately in a realistic, bounded task.
- Use Zero Trust appropriately in a realistic, bounded task.

## WHO THIS IS FOR

Defenders and developers learning lawful, defensive security practice.

## BEFORE YOU BEGIN

- Basic networking and operating-system knowledge

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This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

## COURSE MAP

# From foundation to finished work

Cloud Security - AWS & GCP is a structured, practical course covering AWS, GCP, IAM, Zero Trust. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

| MODULE                              | LESSONS AND FOCUS                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>01 AWS: Security foundations</b> | <ul style="list-style-type: none"><li>1. Assets with AWS</li><li>2. Threats with GCP</li><li>3. Risk with IAM</li></ul>                          |
| <b>02 GCP: Defensive controls</b>   | <ul style="list-style-type: none"><li>1. Identity with Zero Trust</li><li>2. Network protection with AWS</li><li>3. Hardening with GCP</li></ul> |
| <b>03 IAM: Assessment</b>           | <ul style="list-style-type: none"><li>1. Safe testing with IAM</li><li>2. Evidence with Zero Trust</li><li>3. Remediation with AWS</li></ul>     |
| <b>04 Zero Trust: Operations</b>    | <ul style="list-style-type: none"><li>1. Monitoring with GCP</li><li>2. Response with IAM</li><li>3. Recovery with Zero Trust</li></ul>          |

## HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

## CAPSTONE PROJECT

# Produce a reviewable Cloud Security - AWS & GCP project using AWS, GCP, IAM.

|                    |                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------|
| <b>DELIVERABLE</b> | A working Cloud Security - AWS & GCP artefact plus an evidence-based self-review.             |
| <b>TOOLS</b>       | A suitable AWS environment, A plain-text decision log, Test data or realistic sample material |

## Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using AWS.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

## Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

## RESPONSIBLE PRACTICE

Use security techniques only on systems you own or are explicitly authorised to test.

## Evidence log

|                   |                                                      |
|-------------------|------------------------------------------------------|
| <b>INPUT</b>      | Original brief, data, requirements or test material. |
| <b>DECISIONS</b>  | Chosen method and one reasonable alternative.        |
| <b>TESTS</b>      | Normal, boundary and failure results.                |
| <b>LIMITATION</b> | What the result does not establish.                  |
| <b>REVISION</b>   | One evidence-led improvement.                        |

## REVIEW AND SOURCES

# Make the work credible and repeatable

## Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

## Authoritative references

### Cybersecurity Framework 2.0

NIST - reviewed 2026-08-21  
<https://www.nist.gov/cyberframework>

### Web Security Testing Guide

OWASP Foundation - reviewed 2026-08-21  
<https://owasp.org/www-project-web-security-testing-guide/>

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## SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

## NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the AWS scope.